additional abstractions.

Table 4.6 highlights how third-party libraries provide scalable, efficient solutions to common state management challenges in React.

Table 4.6 Challenges in native React state management

Problem	Description	Third-party library solutions
Prop drilling	In React, data is passed down from parent to child components via props. In deeply nested component trees, this can lead to prop drilling, where intermediate components pass props they do not directly use.	Libraries like Zustand, Jotai, or Recoil, allow direct access to global state from any component without passing props through intermediate components.
Global state management	Managing shared state across multiple, unrelated components becomes cumbersome with React's Context API or custom solutions. The Context API can cause performance issues as it triggers re-renders for all components consuming the context, even if only part of the state changes.	Libraries like Redux or MobX centralize application state into a single store, making it easier to manage and debug.
Complexity in scaling	As the application grows, managing state with multiple contexts or reducers becomes harder to maintain and debug. Optimizing re-renders and ensuring compatibility with features like Concurrent	Libraries like Redux are designed to handle complex applications with large teams and codebases by reducing boilerplate and enforcing best practices